CLASS 09

Statistical Tests & Confidence Intervals

GOVT 10: Quantitative Political Analysis

> This chapter turns the logic of inference into specific tools. We run one- and two-sample t-tests and regression t-tests, then introduce confidence intervals, the range of plausible values that pairs every estimate with an honest statement of uncertainty.

1 Statistical Tests

1.1 Sampling Distributions and the Standard Error

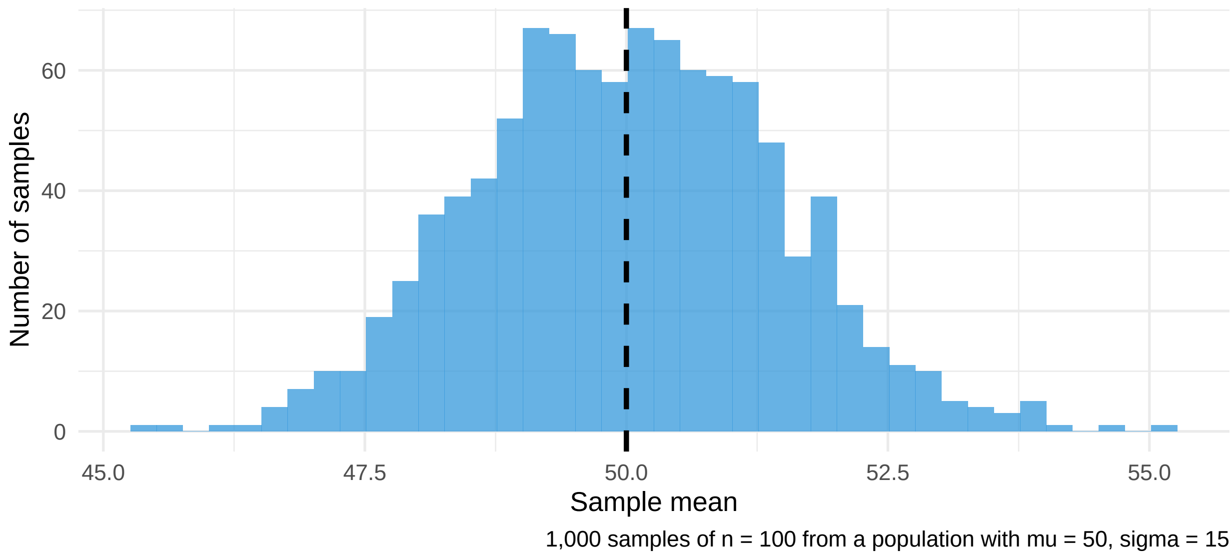
1.1.1 A Sample Is Not the Population

Every statistic in this chapter rests on one uncomfortable fact: we observe a sample, but we care about a population. We compute $\bar{x}$, the mean of the respondents we actually reached, and we want to say something about μ, the mean of everyone we could have reached. These two numbers are never quite equal. Draw a different 1,000 voters and you get a different $\bar{x}$. Nothing has gone wrong when this happens; it is simply what sampling does.

The question inference answers is not "how do we make this gap go away?" — we cannot — but "how large is this gap likely to be?" Once we can put a number on the typical size of the gap, we can say which claims about μ our data can rule out and which it cannot.

1.1.2 The Thought Experiment That Makes Inference Possible

Imagine repeating your study 1,000 times: 1,000 fresh samples of the same size from the same population, each yielding its own $\bar{x}$. You will never actually do this. But you can ask what the collection of those 1,000 means would look like, and that imagined collection — the **sampling distribution** — is the object every test in this chapter is secretly about.



Two features of that picture do all the work. First, the sample means pile up around the true population mean rather than drifting off somewhere else: the sample mean is an unbiased estimator of μ. Second, they form a bell curve, and they do so even when the population itself is nothing like a bell curve. That second fact is the **Central Limit Theorem**, and it is what lets us use the normal and t-distributions on income, turnout, feeling thermometers, and every other lumpy political variable.

1.1.3 Standard Error: Putting a Number on the Spread

The spread of that imagined distribution is the **standard error**. It is estimated from a single sample by

$$SE = \frac{s}{\sqrt{n}}$$

where s is the sample standard deviation and n the sample size. The formula says two things. More variable data give a larger standard error: if individuals differ wildly, any particular sample of them is a shakier guide to the average. Larger samples give a smaller standard error: more observations average out more of the noise.

```r
# How the standard error shrinks with sample size (s held at 15)
tibble(n = c(10, 25, 100, 400, 1600)) %>%
  mutate(se = 15 / sqrt(n))
```

```
# A tibble: 5 x 2
      n    se
  <dbl> <dbl>
1    10 4.74
2    25 3
3   100 1.5
4   400 0.75
5  1600 0.375
```

Notice the shape of that shrinkage. Going from $n = 100$ to $n = 400$ — quadrupling the sample — cuts the standard error in half. Precision improves with the *square root* of sample size, not with sample size itself, which is why the last increment of precision in a survey is so much more expensive than the first. Doubling a sample buys a factor of only about 1.41.

> **Standard deviation is not standard error**
>
> The standard deviation describes how much *individuals* differ from one another. The standard error describes how much a *sample mean* would differ from sample to sample. A survey of 1,000 voters might report an age standard deviation of 15 years and a standard error of 0.47 years. Both are correct, and they answer different questions: the first says ages are spread widely, the second says we have pinned down the average age quite precisely. Reporting one when you mean the other is among the most common errors in undergraduate write-ups.

1.1.4 Three Questions, Three Tools

Everything that follows in this chapter is an answer to one of three questions about that gap between $\bar{x}$ and μ. How far is our result from what we would expect if nothing were going on? That is the **t-statistic**, and its companion the p-value. How precise is our estimate in the first place? That is the **standard error**. What range of values for the true parameter would be consistent with what we saw? That is the **confidence interval**, which occupies the second half of this chapter. The three are not three separate procedures; as the last section will show, they are three views of the same arithmetic.

1.2 The t-Distribution

1.2.1 Gosset's Problem

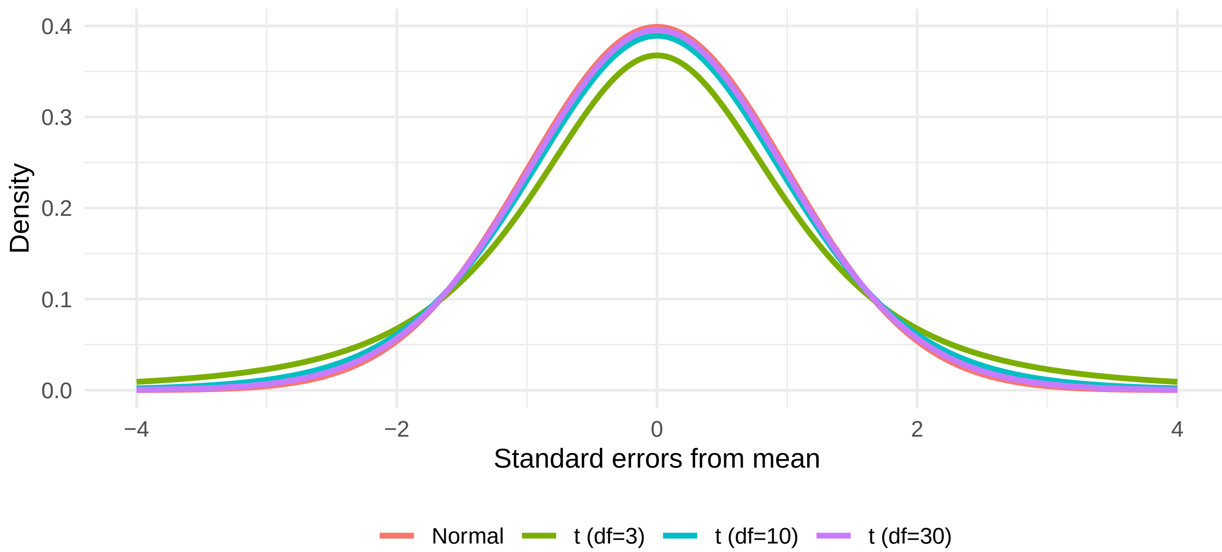
The standard normal distribution describes sampling variability beautifully, but only when we know the population standard deviation σ. In real research we never do. We have a sample, we compute its standard deviation s, and we use that as a stand-in for σ. With a large sample, s is close to σ and nothing much goes wrong. With a small sample, s can be noticeably off, and treating it as though it were the truth makes us overconfident.

This is not a small problem. A reasonable rule of thumb is that with $n < 30$, you should always use the t-distribution and not the normal. With n in the thousands, the two distributions are indistinguishable for practical purposes. Between those extremes, the choice is the t-distribution by default, both because it is always defensible and because modern software computes it without effort.

William Sealy Gosset, a brewer at Guinness around 1908, ran into exactly this problem. Testing quality on small batches of beer, he never had enough observations to estimate σ precisely. He worked out a corrected distribution that accounts for the extra uncertainty of estimating s from limited data. Guinness forbade employees from publishing under their own names, so he used the pseudonym “Student.’’ The result, Student’s t-distribution, is the workhorse of modern inference.

1.2.2 The t-Distribution vs the Normal

The t-distribution looks like the normal but with heavier tails. Heavier tails mean that extreme values are more probable, which is exactly the correction we want when the sample standard deviation might be misleading us. The shape depends on the degrees of freedom: with small n, the tails are noticeably heavy, and as n grows the t-distribution converges to the normal. The intuition is that heavier tails make us less confident in declaring an observation “extreme,’’ which is exactly the right behavior when our estimate of the standard deviation might itself be off.



Heavier tails at small df; converges to normal as df grows

Degrees of freedom equals the sample size minus the number of parameters estimated from the data. For a one-sample t-test, $df = n - 1$ because we used one degree of freedom to estimate the mean. The intuition: if five numbers must average to a known value, only four can be chosen freely; the fifth is determined.

df	Critical value ($\alpha = 0.05$, two-sided)	Implication
5	± 2.571	Small samples need stronger evidence
10	± 2.228	
30	± 2.042	Approaching normal
∞	± 1.960 (normal)	Large-sample standard

Table 1: Critical values shrink as df grows

The practical lesson: with a sample of six, you need a t-statistic above about 2.57 to reach the 0.05 threshold. With a sample of 100, anything above 1.98 will do. Small samples are not just less precise; they require more dramatic evidence to overcome their own noise.

1.3 One-Sample t-Tests for Single Means

1.3.1 When to Use It

The one-sample t-test asks whether a population mean differs from a hypothesized benchmark value. Is the average approval of the president different from 50%? Is the average wait time at a polling place different from the legally mandated 30 minutes? Is the average score on a civic literacy test different from the score we would expect by guessing? Is the average level of trust in elections different from the value reported a decade ago?

In each case there is a benchmark value μ_0 that we want to test against. The benchmark may come from a legal standard, a historical baseline, a theoretical expectation, or a competitor's claim. The one-sample t-test is not a test of whether the mean equals zero; it is a test of whether the mean equals any value of our choosing.

Notice the role of theory here. The benchmark μ_0 does not come from the data; it comes from outside the data. Choosing it well is a matter of substantive judgment. `Approval equals 50\%'' is a defensible benchmark in some contexts (the cutoff between net-positive and net-negative) but not in others.`The civic literacy score equals the random-guessing score'' is a defensible benchmark for testing whether respondents know anything at all, but it is not a benchmark for testing whether respondents are well-informed. The statistics are only as informative as the question you ask of them.

The test statistic is

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where $\bar{x}$ is the sample mean, μ_0 is the hypothesized value under the null, and $s/\sqrt{n}$ is the standard error of the mean. The numerator is the size of the observed gap; the denominator is the size of the gap we would expect from sampling noise. The ratio answers "how many standard errors away from the null is our sample mean?''

1.3.2 Example: Presidential Approval

A national survey of 40 voters yields an average approval rating of about 47.3% with a standard deviation of about 12.1 points. Is this significantly different from 50%?

```r
# One-sample t-test against a benchmark of 50%
approval_test <- t.test(approval_data$approval, mu = 50)
tidy(approval_test)
```

```
# A tibble: 1 x 8
  estimate statistic p.value parameter conf.low conf.high method            alternative
     <dbl>     <dbl>   <dbl>     <dbl>    <dbl>     <dbl> <chr>             <chr>
1     48.8    -0.578   0.567        39     44.5      53.1 One Sample t-test two.sided
```

> Read the output column by column. `estimate` is the sample mean. `statistic` is the t-statistic — the number of standard errors between $\bar{x}$ and 50. `p.value` is the probability of

> seeing a t-statistic this far from zero (in either direction) if the truth were 50. `conf.low` and `conf.high` give the 95% confidence interval, which contains plausible values for the true mean.

In this run, the sample mean lands just below 47 and the t-statistic is roughly -1.5. The p-value is well above 0.05, so we fail to reject the null that approval equals 50%. Notice the careful phrasing: we have not proven approval is exactly 50%. We have failed to find evidence that it differs.

1.3.3 Reading the Result

A few things deserve attention. First, the confidence interval includes 50, which is geometrically equivalent to the p-value exceeding 0.05. The connection between intervals and tests is exact: if a two-sided test at level α fails to reject the null value μ_0, then the corresponding $(1 - \alpha)$ confidence interval contains μ_0, and vice versa. This duality is one of the most useful features of the framework: a confidence interval communicates both an estimate and the result of a hypothesis test at the same time.

Second, the sample size is small enough that we are using the t-distribution rather than the normal, which costs us slightly in precision. Compare the critical values: with $df = 39$ we need $|t| > 2.02$ to reject at the 0.05 level, whereas with the normal distribution we would have needed only $|t| > 1.96$. The difference is small here but grows for smaller samples.

Third, the substantive conclusion changes nothing about practice: we report the estimate, the standard error, and the p-value, and let readers judge. A formal test gives a binary decision; the underlying numbers tell a richer story. A polite reading of these results might say "the sample mean of 47.5% is consistent with population approval of 50%, but our estimate is too imprecise to say much beyond that.''

1.4 Two-Sample t-Tests for Comparing Groups

1.4.1 The Workhorse of Empirical Research

Most political-science questions boil down to a comparison between two groups. Did the treatment group turn out at a higher rate than the control group? Do Democrats and Republicans differ in political knowledge? Do battleground states have higher participation than safe states? Do voters who consume news regularly differ from those who do not in their assessment of presidential job performance? The two-sample t-test addresses all of these.

Even more importantly, the two-sample t-test is the inferential half of the experimental designs we studied in the previous chapter. After random assignment, the t-test compares average outcomes in treatment and control. The two halves of the experimental method are random assignment (which gives us a causal interpretation of differences) and the two-sample t-test (which tells us whether those differences are too large to attribute to chance).

The null is that the two population means are equal, $\mu_1 = \mu_2$. The alternative is that they differ.

The test statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}.$$

The numerator is the observed difference in means. The denominator is the standard error of that difference, which combines variability from both groups.

> **Common mistake: averaging the SEs**
>
> The standard error of a difference is built from variances, not from standard errors directly. The variances of the two group means add; we then take the square root. "Averaging the SEs" is the wrong recipe and will give the wrong answer.

A useful intuition is that the standard error of a difference is always at least as large as either group's individual standard error, and often larger. We are combining two sources of uncertainty, not averaging them away. This is why even quite large between-group differences can fail to reach significance with modest sample sizes, and why doubling the sample size in each group is more useful than doubling it in just one group.

1.4.2 Example: Door-to-Door GOTV Experiment

A get-out-the-vote experiment randomly assigns 500 registered voters to either receive a door knock from a canvasser or be left alone. After the election, we record who voted.

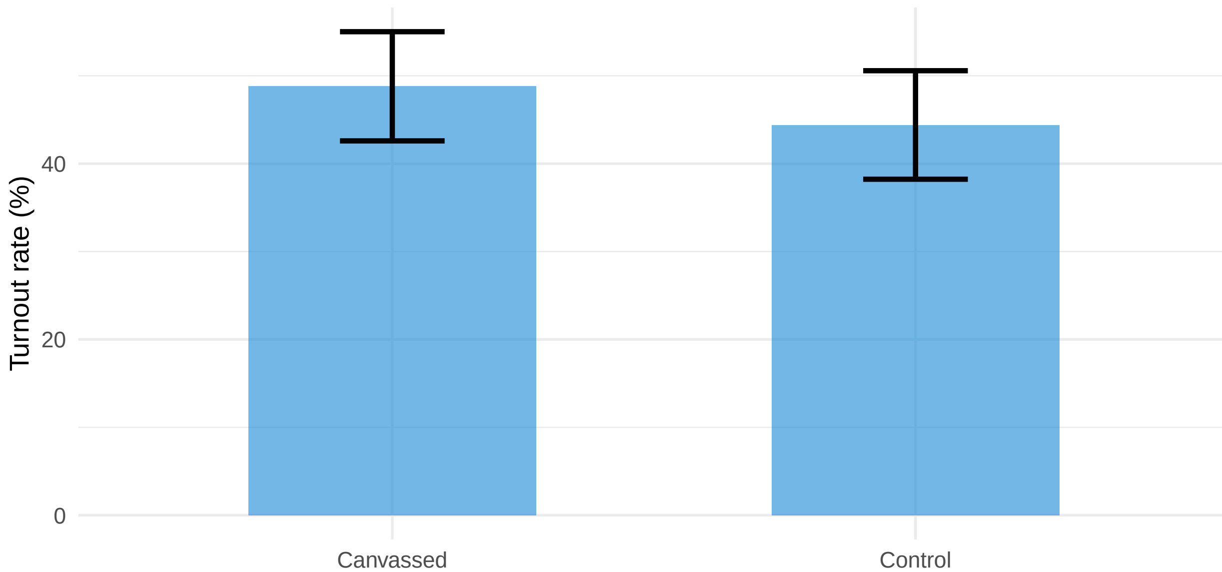
```r
# Group-level turnout rates
gotv %>%
  group_by(treatment) %>%
  summarise(
    n = n(),
    turnout_rate = mean(turnout_pct),
    sd = sd(turnout_pct),
    se = sd / sqrt(n)
  )
```

```
# A tibble: 2 x 5
  treatment     n turnout_rate    sd    se
  <chr>     <int>        <dbl> <dbl> <dbl>
1 Canvassed   250         48.8  50.1  3.17
2 Control     250         44.4  49.8  3.15
```

```r
# Two-sample t-test
gotv_test <- t.test(turnout_pct ~ treatment, data = gotv)
tidy(gotv_test)
```

```
# A tibble: 1 x 10
  estimate estimate1 estimate2 statistic p.value parameter conf.low conf.high method
     <dbl>     <dbl>     <dbl>     <dbl>   <dbl>     <dbl>    <dbl>     <dbl> <chr>
1     4.40      48.8      44.4     0.985   0.325      498.    -4.38      13.2 Welch Two S~
# i 1 more variable: alternative <chr>
```

> The tilde formula `turnout_pct ~ treatment` tells R to compare turnout across the two values of `treatment`. `estimate1` and `estimate2` are the two group means; `estimate` is their difference. The reported confidence interval is on the *difference*, not on either group separately.



The point estimates suggest canvassing increases turnout by a few percentage points, but whether the difference clears the bar for significance depends on the sample. The overlap of the confidence intervals tracks the p-value: heavily overlapping intervals correspond to a non-significant test.

In real GOTV experiments, the famous Gerber-Green door-knocking studies of the early 2000s found canvassing effects on the order of seven to ten percentage points. Detecting an effect that size with high confidence required samples in the thousands. With a few hundred voters, even substantial effects can hide inside the standard error. This is a recurring lesson in field experimentation: the size of an effect that you can statistically detect depends on the size of your study.

1.4.3 Example: Gender Gap in Policy Support

This is the same gender-gap scenario sketched in last week's chapter, now worked through end-to-end with code and a real t-test. Treat the two as the same scenario in two formats: a conceptual sketch first, the full t-test here.

A nationally representative comprehensive opinion survey asks 400 respondents (200 men and 200 women) whether they support stricter gun laws.

```
# Compare support across gender
gun_survey %>%
  group_by(gender) %>%
  summarise(
    n = n(),
```

```
    support = mean(supports_strict_laws),
    se = sqrt(support * (1 - support) / n)
  )
```

```
# A tibble: 2 x 4
  gender      n support     se
  <chr>   <int>   <dbl>  <dbl>
1 Men       200   0.51  0.0353
2 Women     200   0.685 0.0328
```

```
# Two-sample t-test on the binary outcome
gender_gap_test <- t.test(supports_strict_laws ~ gender, data = gun_survey)
tidy(gender_gap_test) %>% select(estimate, statistic, p.value, conf.low, conf.high)
```

```
# A tibble: 1 x 5
  estimate statistic  p.value conf.low conf.high
     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1   -0.175     -3.62 0.000336   -0.270   -0.0799
```

> A ten-point observed gap with standard errors of about three points in each group works out to a roughly two-standard-error difference, comfortably below the conventional 0.05 threshold. We have evidence that men and women in the population differ in their support for the policy.

1.4.4 When Two-Sample t-Tests Misbehave

The two-sample t-test is robust to moderate departures from its assumptions, but a few warnings are in order. Extreme outliers can drag means and inflate variances. Heavily skewed distributions with very small samples can mislead. Very unequal variances are mostly handled by R's default Welch correction, which does not assume equal variances between groups.

> **Practical guidance**
>
> With $n \geq 30$ in each group, t-tests work well even when individual observations are not normally distributed. With smaller groups, look at the data first. A boxplot or histogram catches outliers and skew that a numerical summary would hide.

A related issue is the assumption of independence between observations. The standard t-test formulas assume that each observation is sampled independently of the others. In practice, voters within the same household, students within the same school, or respondents recruited through the same network may share characteristics that violate independence. Ignoring this clustering tends to make standard errors look smaller than they should be, which inflates t-statistics and shrinks p-values misleadingly. Modern survey research uses methods like clustered standard errors and survey weights to handle these complications, but the basic t-test is the foundation on which they are built.

A final issue worth flagging: when the outcome is binary (turnout, support, vote choice), a two-sample t-test compares proportions and is closely related to a chi-squared test of independence. R's `t.test()` will run on a 0/1 outcome and produce sensible results for moderately sized samples. With very small samples or rare outcomes, more specialized tests (such as Fisher's exact test) become preferable.

1.5 Regression t-Tests for Coefficients

1.5.1 Every Coefficient is a Hypothesis Test

When R reports a regression, each row of the coefficient table comes with a t-statistic and a p-value. These are not decoration. Every coefficient in a regression carries an automatic hypothesis test: the null is that the population coefficient equals zero (no relationship between this predictor and the outcome), and the alternative is that it does not.

The test statistic is

$$t = \frac{\hat{\beta} - 0}{SE(\hat{\beta})} = \frac{\hat{\beta}}{SE(\hat{\beta})}.$$

This is structurally identical to a one-sample t-test: the estimate over its standard error. If $|t|$ is large enough — roughly above 2 for typical sample sizes — the coefficient is significantly different from zero.

The fact that R computes a t-statistic and p-value for every coefficient means that a regression table with five predictors is, simultaneously, five separate hypothesis tests. This is convenient, but it raises a version of the multiple-comparisons concern we will return to later: if you fit a regression with twenty predictors, you should not be surprised when one of them comes back with $p < 0.05$ by chance.

1.5.2 Example: Campaign Spending and Vote Share

Consider a sample of 150 congressional candidates. For each, we observe campaign spending (in thousands), whether the candidate is the incumbent, and vote share.

```r
# Regression with two predictors
model <- lm(vote_share ~ campaign_spending + incumbent, data = campaign)
tidy(model)
```

```
# A tibble: 3 x 5
  term              estimate std.error statistic  p.value
  <chr>                <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)        27.4      1.11         24.7 3.39e-54
2 campaign_spending   0.0259   0.00140      18.5 3.98e-40
3 incumbent           8.01     0.763        10.5 1.39e-19
```

> Each row of the coefficient table is a separate test. `estimate` is $\hat{\beta}$. `std.error` is the standard error of that estimate. `statistic` is the t-statistic ($\hat{\beta}/SE$). `p.value` is the probability of a t-statistic this far from zero if the true coefficient were exactly zero.

The campaign-spending coefficient is small in absolute terms (each additional thousand dollars buys a fraction of a percentage point) but highly significant because the standard error is even smaller. The incumbency coefficient is large and also highly significant. The interpretation in multiple regression is conditional: each coefficient describes the effect of its variable *holding the others constant*.

Comparing the same coefficient in a simple regression and in a multiple regression often produces different numbers. In a simple regression of vote share on campaign spending, the spending coefficient absorbs the part of vote share that incumbency would otherwise explain. In a multiple regression that includes incumbency, the spending coefficient describes spending's effect *after* we have controlled for incumbency. This is the same conditional interpretation we developed in Chapter 7, now connected to a formal hypothesis test.

1.5.3 What Determines the Standard Error of a Coefficient

Three things drive the precision of a regression coefficient. Larger samples shrink the standard error in proportion to $1/\sqrt{n}$. More residual variance (more scatter around the regression line) inflates the standard error. More variance in the predictor itself shrinks the standard error, because a predictor that takes many different values gives more leverage to detect a relationship.

```r
# A small simulation to demonstrate sample size effect on SE of a coefficient
set.seed(2024)
se_by_n <- map_dfr(c(50, 200, 800), function(n) {
  betas <- replicate(500, {
    x <- rnorm(n, mean = 50, sd = 15)
    y <- 20 + 0.8 * x + rnorm(n, mean = 0, sd = 10)
    coef(lm(y ~ x))[2]
  })
  tibble(n = n, se_empirical = round(sd(betas), 4))
})
se_by_n
```

```
# A tibble: 3 x 2
      n se_empirical
  <dbl>        <dbl>
1    50       0.0973
2   200       0.0487
3   800       0.0237
```

> Quadrupling n from 50 to 200, and again from 200 to 800, roughly halves the empirical standard error each time. This is the same $1/\sqrt{n}$ shrinkage we saw for sample means: regression is just running the same machinery on a more complicated target.

The practical lesson is that not every variable in a regression carries the same evidentiary weight. A coefficient might be small in absolute size but highly significant because it is measured precisely; another might be large but insignificant because it is measured noisily. Pair each coefficient with its standard error before reading too much into its sign.

A related connection worth noting is that the two-sample t-test and a regression on a single binary predictor produce the same t-statistic and the same p-value. The two procedures are not just similar; they are mathematically identical. Recognizing this connection makes regression less mysterious: it is the same hypothesis-testing machinery we built in the previous section, generalized to allow more than two groups and to control for additional variables.

> **Two-sample t-test = regression with a binary predictor**
>
> The two-sample t-test and a regression of the outcome on a 0/1 indicator for group membership yield the same t-statistic and p-value. Recognizing this connection makes regression less mysterious: it is the same hypothesis testing machine, generalized.

1.6 Common Pitfalls in Hypothesis Testing

The mechanics of testing are now in place. The hardest part of inference is not running the tests but avoiding the predictable misuses. Three pitfalls deserve careful attention.

The three pitfalls below all share a common structure: each describes a way in which the nominal 5% false-positive rate of a single test can be silently inflated through researcher choices that may not even feel dishonest. Vigilance about these patterns is what separates routine statistical practice from rigorous statistical practice.

1.6.1 Multiple Comparisons

If you test 20 hypotheses at $\alpha = 0.05$, the laws of probability guarantee that on average one will appear significant by chance alone, even if every null hypothesis is true.

```r
# Run 20 tests on completely null data and see how many "find" something
set.seed(2026)
null_results <- map_dbl(1:20, function(i) {
  group_a <- rnorm(100, mean = 50, sd = 10)
  group_b <- rnorm(100, mean = 50, sd = 10)   # truly identical populations
  t.test(group_a, group_b)$p.value
})

cat("Number of significant results out of 20 tests:",
    sum(null_results < 0.05), "\n")
```

```
Number of significant results out of 20 tests: 0
```

```r
cat("Smallest p-value:", round(min(null_results), 4))
```

```
Smallest p-value: 0.0535
```

> Every one of these 20 tests compared groups that came from identical distributions. The null hypothesis was true in every case. By chance, about one of them looks “significant.” If a researcher reports only the significant one and stays quiet about the other nineteen, the resulting paper is misleading.

The defense against multiple comparisons is to pre-register hypotheses, report all tests, and adjust thresholds when running many tests. Several techniques (Bonferroni, false discovery rate) correct for multiple testing formally; the simplest defense is honesty about the number of comparisons run.

1.6.2 P-Hacking

P-hacking is a family of behaviors that turn null findings into significant ones through analytical flexibility. Examples include trying many model specifications and reporting the one that works,'' adding observations until p drops below 0.05, dropping observations until it does, choosing the outcome variable that gives the best p-value, and changing the hypothesis after seeing the data (sometimes called HARKing,hypothesizing after results are known'').

The cumulative effect of these practices is a literature in which published findings replicate at much lower rates than they should. The fix is largely cultural: pre-register designs, share data and code, report effect sizes alongside p-values, and acknowledge null results.

P-hacking is often unintentional. A researcher tries one specification, sees an interesting result, tries another, and another, and eventually publishes the one that crossed the threshold. Each individual choice felt defensible at the time. The cumulative effect, summed across the entire research community, looks like a literature riddled with false positives. The most reliable protection against p-hacking is to write down a single analysis plan before the data are collected and stick to it.

> **Common mistake: stopping data collection when $p < 0.05$**
>
> If you peek at your data repeatedly and stop the moment the test crosses the threshold, the nominal Type I error rate of 5% climbs much higher. Decide your sample size in advance and stick to it.

1.6.3 “Almost Significant''

You will encounter, in journal articles and in the popular press, results described as approaching significance,''marginally significant,'' or “trending toward significance.'' These phrases dress up the fact that p landed above 0.05 in language that sounds more impressive.

The 0.05 threshold is arbitrary. There is no honest sense in which $p = 0.049$ is fundamentally different from $p = 0.051$. The right response is to report the exact p-value, place it in context with effect size and sample size, and let the reader judge. The wrong response is to invent linguistic categories that try to claim significance without earning it.

A related rhetorical move worth recognizing is the use of one-sided tests after the fact. A two-sided test asks whether two groups differ; a one-sided test asks whether one specific group is larger than the other. Switching from two-sided to one-sided after seeing the data effectively halves the p-value and can flip a borderline result into significance. Choose the direction of your test before you look at your data, or stick with two-sided tests by default.

A subtler variant of the same problem is selective rounding. Reporting $p = 0.04$ when the actual computed value is 0.048 obscures the fact that the result is on the borderline. Software returns p-values to many decimal places; honesty demands reporting at least two or three. When in doubt, report the exact value to three decimal places and let readers form their own opinion about whether the evidence is strong, weak, or borderline.

> **Better reporting**
>
> A complete report of a finding includes the effect size, the sample size, the p-value or confidence interval, and a substantive interpretation. "Women supported the policy 12 points more than men (65% vs 53%; $n = 400$; $p = 0.03$)" is informative. "The difference was significant ($p < 0.05$)" is not.

1.6.4 Failing to Reject is Not Accepting

When $p > 0.05$, we say we `fail to reject $H_0$,'' not that we`accept H_0.'' The clumsy phrasing is deliberate. A non-significant result could mean the null is true, or that there is a real effect but our sample lacked the power to detect it. Distinguishing these requires a power analysis or a confidence interval narrow enough to rule out interesting effect sizes.

A study that fails to find an effect with a small sample is almost uninformative about whether the effect exists. A study that fails to find an effect with a large, well-powered sample is genuine evidence of a small or absent effect. The two interpretations look identical in a journal abstract but mean very different things.

The same logic applies in reverse: a study that finds an effect with a very large sample is showing that the effect is not exactly zero, but it is not necessarily showing that the effect is large enough to matter. Statistical significance is a coarse instrument that gets coarser as samples grow.

```r
# Demonstration: a real 3-point effect with low power often goes undetected
set.seed(8)
detection_rate <- mean(replicate(1000, {
  control   <- rnorm(50, mean = 50, sd = 12)
  treatment <- rnorm(50, mean = 53, sd = 12)   # real 3-point effect
  t.test(control, treatment)$p.value < 0.05
}))
cat("Detection rate with n=50/group:", round(detection_rate, 2))
```

```
Detection rate with n=50/group: 0.25
```

> A real three-point effect goes undetected most of the time when we only have 50 observations per group. Concluding "no effect" from such a study would be a mistake. Failing to find evidence is not the same as finding evidence against.

2 Confidence Intervals

2.1 From Hypothesis Tests to Plausible Ranges

In the previous class we learned to ask a sharp yes-or-no question: is the effect significantly different from zero? We computed a t-statistic, looked up a p-value, and either rejected the null hypothesis or failed to reject it. That framework is powerful, but it is also narrow. It tells us whether evidence exists against the null. It does not tell us how big the effect is or how precisely we have measured it.

Consider a get-out-the-vote experiment. A canvassing campaign increased turnout by 4 percentage points with p = 0.03. The result is statistically significant. But should the campaign manager invest another million dollars next cycle? If the true effect could plausibly be as small as 0.5 points, the program may not justify its cost. If the true effect could plausibly be as large as 8 points, it could decide a close election. The p-value cannot distinguish these scenarios. We need a range.

Confidence intervals address this directly. Instead of reporting just "4 points (p = 0.03)," we report "4 points, 95% CI [0.5, 7.5]." That one expression communicates the point estimate, the range of plausible truth, and the significance status all at once. The interval excludes zero, so we know p < 0.05 without doing a second calculation.

> **Why CIs Matter**
>
> P-values answer one question: how surprising is this result if nothing is going on? Confidence intervals answer a richer question: what values of the underlying truth are consistent with what we observed? Polls in elections are reported as "54% support, margin of error $\pm$3 points" precisely because a range is more informative than a point estimate alone.

2.2 The Building Blocks: t-Statistics Become Intervals

Earlier you computed a t-statistic as the coefficient divided by its standard error:

$$t = \frac{\hat{\beta}}{SE(\hat{\beta})}$$

We then asked: is this t large enough to reject the null? Equivalently, we asked whether $\hat{\beta}$ is far enough from zero in standard error units to be implausible under H_0.

A confidence interval flips this question around. Instead of asking whether zero is too far from our estimate, we ask: what range of values is within "reasonable distance" of our estimate? For a 95% CI, "reasonable distance" is the critical value t^* from the t-distribution that captures the middle 95% of the curve.

$$\text{CI} = \hat{\beta} \pm t^* \times SE(\hat{\beta})$$

This formula has three ingredients you already know. The point estimate $\hat{\beta}$ is your best guess from the sample. The critical value t^* comes from `qt()` and depends on the desired confidence level and the degrees of freedom. The standard error SE is the same one used to compute the t-statistic in a hypothesis test.

```r
# Critical values for 95% confidence at different sample sizes
sample_sizes <- c(40, 100, 500, 10000)
critical_values <- tibble(
  n = sample_sizes,
  df = n - 2,
  t_critical = round(qt(0.975, df = df), 4)
)
critical_values
```

```
# A tibble: 4 x 3
      n    df t_critical
  <dbl> <dbl>      <dbl>
1    40    38       2.02
2   100    98       1.98
3   500   498       1.96
4 10000  9998       1.96
```

> Notice that the critical value shrinks toward 1.96 as the sample grows. With a small sample (n = 40), we need a wider net (t* = 2.02) to be 95% confident. With a very large sample, the t-distribution becomes indistinguishable from the normal and we recover the familiar 1.96 multiplier. This is the same Student's t-distribution from the previous class doing the same job in a new costume.

2.3 Constructing a CI by Hand

Suppose a small randomized canvassing study reports a coefficient of $\hat{\beta} = 0.82$ for the effect of contacts on precinct turnout, with $SE = 0.25$ and $n = 40$. We want a 95% CI.

Walking through it in prose first:

- **Step 1.** The point estimate is $\hat{\beta} = 0.82$.
- **Step 2.** With $df = n - 2 = 38$, the critical value t^* for a 95% interval is about 2.02 (look up in a t-table or call `qt(0.975, 38)`).
- **Step 3.** The margin of error is $t^* \times SE = 2.02 \times 0.25 \approx 0.51$.
- **Step 4.** The 95% CI is $\hat{\beta} \pm$ margin $= 0.82 \pm 0.51$, or roughly $[0.31, 1.33]$.

Now the same thing in code, so you can swap in different numbers:

```r
# Step 1: ingredients
beta_hat <- 0.82
se       <- 0.25
n        <- 40

# Step 2: critical value (df = n - 2 for simple regression)
t_crit <- qt(0.975, df = n - 2)

# Step 3: margin of error
margin <- t_crit * se

# Step 4: bounds
ci_lower <- beta_hat - margin
```

```r
ci_upper <- beta_hat + margin

cat("95% CI: [", round(ci_lower, 2), ",", round(ci_upper, 2), "]")
```

```
95% CI: [ 0.31 , 1.33 ]
```

> The interval runs from roughly 0.31 to 1.33. Three plausible-truth statements follow. First, zero is not in the interval, so we can already announce that the effect is significant at $\alpha = 0.05$. Second, the truth could easily be as small as 0.3 contacts per percentage point or as large as 1.3—a four-fold range—so the precision is modest. Third, the point estimate of 0.82 remains our single best guess, but it is far from the only value consistent with these data.

2.4 Doing It the Easy Way: `t.test()`, `confint()`, and `broom::glance()`

R hands you confidence intervals automatically. The three workhorse functions covered in lecture are `t.test()`, `confint()`, and `broom::glance()`.

```r
# A pre-election poll: is support different from 50%?
test_result <- t.test(poll_data$supports, mu = 0.50)
tidy(test_result)
```

```
# A tibble: 1 x 8
  estimate statistic   p.value parameter conf.low conf.high method           alternative
     <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl> <chr>            <chr>
1    0.569      3.92 0.0000946       799    0.534     0.603 One Sample t-test two.sided
```

> **Reading the `tidy()` output line by line.**
>
> **1. `estimate`.** Sample support is about 54.5%—our best single guess.
>
> **2. `conf.low` and `conf.high`.** The 95% CI runs from roughly 51.0% to 58.0%. These are the plausible values for true support.
>
> **3. Does the CI exclude 50?** Yes. We reject the null that support equals 50%.
>
> **4. `p.value`.** Below 0.05, confirming what the CI already told us.
>
> **5. `statistic`.** The t-statistic, well above 2 in absolute value.
>
> **Template sentence:** ”Support for the initiative (54.5%, 95% CI [51.0%, 58.0%]) was significantly above 50%, t(799) = 2.55, p = 0.011.”

For regression, `confint()` returns the bounds for each coefficient, and `broom::tidy(model, conf.int = TRUE)` returns the same information in a tidy data frame ready for plotting.

```r
# Vote share regression
model <- lm(vote_share ~ ad_spending + incumbent, data = campaign_data)

# Two ways to get CIs
confint(model)
```

```
                  2.5 %      97.5 %
(Intercept) 32.82340759 38.79080097
```

```
ad_spending  0.01688771  0.02816719
incumbent    7.48251157 11.38922172
```

```r
tidy(model, conf.int = TRUE)
```

```
# A tibble: 3 x 7
  term        estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)  35.8      1.51        23.7  1.68e-59  32.8      38.8
2 ad_spending   0.0225   0.00286      7.88 2.21e-13   0.0169    0.0282
3 incumbent     9.44     0.991        9.53 6.19e-18   7.48     11.4
```

> The `tidy()` output stacks the estimate, standard error, t-statistic, p-value, and the 95% CI bounds in one tibble per coefficient. The `confint()` output is leaner: just the lower and upper bounds. Use whichever fits the task. `broom::glance()` (not shown) returns one-row whole-model summaries—R^2, RMSE, and degrees of freedom—useful when you want to compare overall model fit.

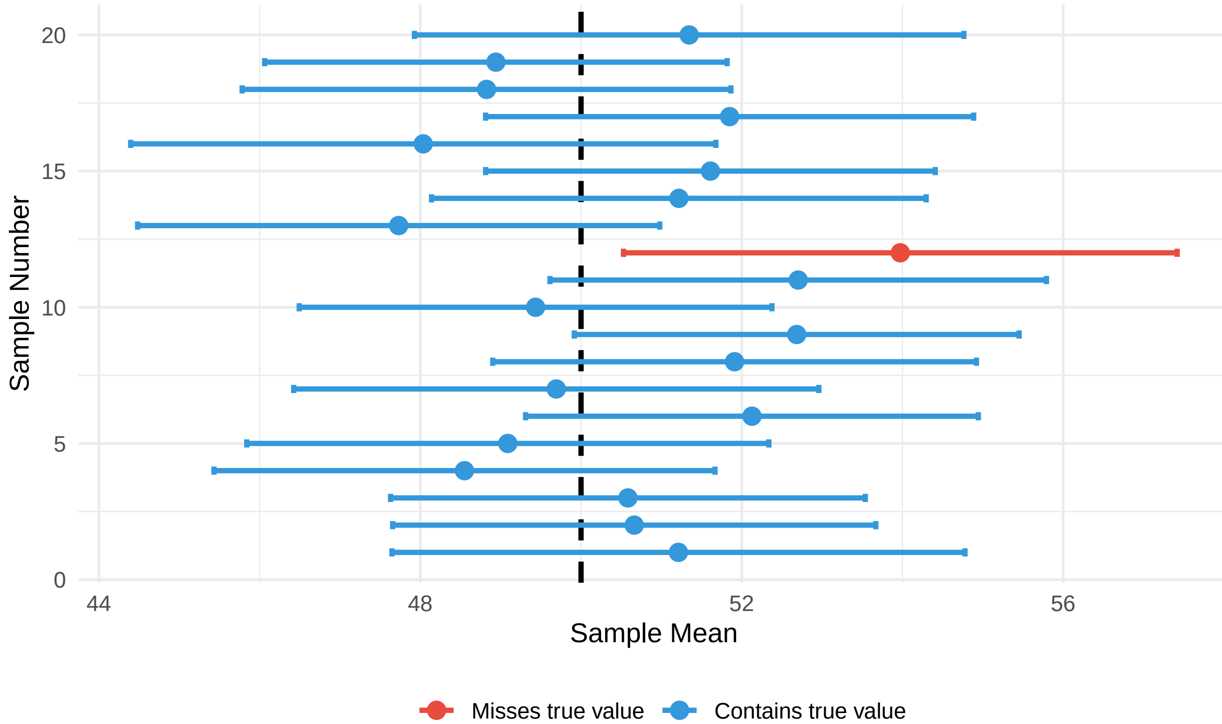
2.5 What "95% Confidence" Actually Means

The single most-tested concept on the final is the correct interpretation of a confidence interval. The statement "95% confidence" refers to the long-run behavior of the procedure that built the interval, not to the probability that the truth lies in any one specific interval.

Correct. If we repeated our sampling procedure many times, building a new 95% CI each time, roughly 95% of those intervals would cover the true parameter.

Incorrect. "There is a 95% probability the true value is between 51% and 58%."

Why is the second statement wrong? Because the true parameter is fixed. It is either inside our interval or it is not. The randomness in the construction comes entirely from the sample we drew. Probability statements apply to the procedure, not to the unobservable truth.



erval is one of the ~5% that miss the truth --- these always exist, and we can't know whether our own sample is one of them.

Each horizontal bar is a confidence interval built from a different random sample drawn from the same population. Most intervals capture the true mean (50, shown as the dashed line). The few red ones miss entirely — they are one of the roughly 5% of intervals that always exist by construction, and we have no way of knowing whether our particular interval is one of them. In the long run, roughly 95% succeed—which is what the "95%" describes.

> **Common Mistake: Probability Interpretation**
>
> Do not say "there's a 95% probability the true value is between 51% and 58%." Say instead "we are 95% confident the true value is between 51% and 58%," understanding that this confidence refers to the procedure's long-run reliability, not to the probability that any particular interval has captured the truth.

2.6 More Examples of Correct vs. Incorrect Interpretations

Because this concept is so frequently mishandled, it is worth working through several political scenarios.

Example 1: Election poll. A pre-election poll of 800 likely voters finds 54% support for the incumbent. The 95% CI is [51%, 57%].

Statement	What it does	Verdict
”There’s a 95% chance true support is between 51% and 57%.”	Treats the parameter as random	**Incorrect**
”95% of voters support the incumbent at a level between 51% and 57%.”	Confuses CI with population spread	**Incorrect**
”If we ran this poll many times, 95% of CIs would contain true support.”	Describes the procedure	**Correct**
”51%–57% are plausible values for true support given these data.”	Informal but accurate	**Correct**

Example 2: Treatment effect. A randomized debate-watching experiment finds a 0.8-point increase in political knowledge with 95% CI [0.3, 1.3].

Incorrect Statement	Why it’s wrong
”We are 95% sure the effect is exactly 0.8 points.”	0.8 is the point estimate regardless of confidence level; the CI is about uncertainty, not the estimate itself
”95% of participants improved by 0.3 to 1.3 points.”	The CI describes the average effect, not individual outcomes; individuals vary much more
”The probability the effect is positive is 95%.”	The CI being entirely positive tells us the effect is likely positive at $\alpha = 0.05$, but 95% refers to the procedure’s reliability, not the probability of a positive effect

> The key distinction is always between statements about the parameter (which is fixed) and statements about the procedure (which is random across samples). The truth does not move. What moves is your interval, sample to sample.

2.7 The Connection to Hypothesis Tests

A 95% confidence interval and a hypothesis test at $\alpha = 0.05$ carry the same information. They use the same standard error, the same t-distribution, and the same logic. The CI excludes zero if and only if the p-value is below 0.05.

If you know this...		You also know this
95% CI excludes 0	$\Leftrightarrow$	p < 0.05 (significant)
95% CI includes 0	$\Leftrightarrow$	p $\geq$ 0.05 (not significant)
99% CI excludes 0	$\Leftrightarrow$	p < 0.01
90% CI excludes 0 but 95% CI includes 0	$\Leftrightarrow$	0.05 < p < 0.10

Table 2: Two-way translation between CIs and p-values

This is a practical reading tool. If a journal article reports only confidence intervals, you can extract significance information without redoing any math. If a published table reports only p-values, you know whether the 95% CI crosses zero.

> **Concrete pair.** A study reports an estimated effect of 1.25 with a 95% CI of [0.5, 2.0]. Because the interval excludes 0, we know *without seeing the p-value* that $p < 0.05$. Conversely, if a study reports $\hat{\beta} = 1.25$ with $p = 0.01$, we know the 95% CI excludes 0.

```r
# Demonstrate the equivalence
test <- t.test(outcome ~ group, data = treatment_effect)
cat("P-value:", round(test$p.value, 4), "\n")
```

```
P-value: 0
```

```r
cat("95% CI for difference:", round(test$conf.int[1], 2), "to",
    round(test$conf.int[2], 2), "\n")
```

```
95% CI for difference: -13 to -5.25
```

```r
cat("CI excludes zero?", test$conf.int[2] < 0 | test$conf.int[1] > 0, "\n")
```

```
CI excludes zero? TRUE
```

```r
cat("p < 0.05?", test$p.value < 0.05)
```

```
p < 0.05? TRUE
```

2.8 Precision: Narrow vs. Wide Intervals

A confidence interval communicates *how much* you know, not just *what* you know. Narrow intervals mean precise estimates. Wide intervals mean fuzzy ones. Two studies can report the same point estimate but provide vastly different evidence.

```r
# Same true effect, two sample sizes
small_ci <- confint(lm(y ~ x, data = small))["x", ]
big_ci   <- confint(lm(y ~ x, data = big))["x", ]

cat("Small sample (n = 40) 95% CI:", round(small_ci, 3), "\n")
```

```
Small sample (n = 40) 95% CI: 0.303 0.638
```

```r
cat("Large sample (n = 500) 95% CI:", round(big_ci, 3))
```

```
Large sample (n = 500) 95% CI: 0.501 0.596
```

> Both samples are drawn from the same underlying population with a true slope of 0.5, but the small-sample CI is several times wider than the large-sample CI. Quadrupling the sample size roughly halves the width, since the standard error scales as $1/\sqrt{n}$. To halve the width of a CI, you need to quadruple the sample.

Three factors determine precision. First, sample size: bigger samples give narrower CIs through the $1/\sqrt{n}$ relationship. Second, residual variability: noisier outcomes give wider CIs. Third, in regression, the spread of the predictor variable: wider X spread gives narrower CIs because the slope is pinned down by leverage at the extremes.

2.9 Higher Confidence Costs Width

To be more confident that our interval covers the truth, we have to cast a wider net. There is no free lunch.

```r
# Three confidence levels on the same data
sample_mean <- 52
sample_se   <- 2.5
n           <- 40

levels <- c(0.90, 0.95, 0.99)
ci_compare <- tibble(
  confidence = paste0(levels * 100, "%"),
  t_critical = round(qt((1 + levels) / 2, df = n - 1), 3),
  lower  = round(sample_mean - t_critical * sample_se, 2),
  upper  = round(sample_mean + t_critical * sample_se, 2),
  width  = round(upper - lower, 2)
)
ci_compare
```

```
# A tibble: 3 x 5
  confidence t_critical lower upper width
  <chr>           <dbl> <dbl> <dbl> <dbl>
1 90%              1.68  47.8  56.2  8.42
2 95%              2.02  46.9  57.1 10.1
3 99%              2.71  45.2  58.8 13.5
```

> Moving from 90% to 99% confidence inflates the interval width from about 8.4 to 13.8 points. Most political research uses 95%. Some clinical trials use 99% when a wrong conclusion would be costly. Use what your audience expects unless you have a good reason to deviate.

2.10 Communicating Effect Size AND Uncertainty

A common failure mode in undergraduate write-ups is to report the point estimate and stop. A better write-up reports the estimate, its CI, and a substantive interpretation in plain language.

Weak. "Phone banking had a significant effect on turnout (p = 0.02)."

Better. "Phone banking increased turnout by 7 percentage points (p = 0.02)."

Strong. "Phone banking increased turnout by 7 percentage points (95% CI: 1.2 to 12.8, p = 0.02, n = 400). This effect is comparable to the boost from a competitive Senate race on the same ballot and would shift outcomes in close districts."

The strong version tells a complete story: direction, magnitude, uncertainty, sample size, and substantive scale. The reader can decide whether the effect is large enough to justify the program.

> **Reporting Template**
>
> 1. Direction and significance: "X was positively / negatively / not significantly associated with Y."
> 2. Statistics: "$\hat{\beta}$ = ___, 95% CI [___, ___], t(df) = ___, p = ___."
> 3. Substantive scale: "Each unit increase in X is associated with a ___-unit change in Y."
> 4. Precision: "The interval is narrow/wide, suggesting high/low precision."
> 5. Practical meaning: compare to a benchmark the reader will recognize.

2.11 Common Pitfalls

Pitfall 1: Statistical vs. practical significance. With n = 10,000, a 0.3-point change on a 100-point scale can be statistically significant. That does not mean it matters. Always evaluate effect size against a substantive benchmark.

Pitfall 2: Non-significance is not "no effect." A wide CI that crosses zero usually means the study lacked power, not that the truth is zero. Treat "fail to reject" as "we don't know yet," not "nothing to see."

Pitfall 3: Overlapping CIs and significance. Two groups can have overlapping CIs while the difference between their means is still significant. Always test the difference directly with `t.test()` rather than eyeballing error bars — the difference can be significant even when the intervals appear to overlap, because the SE of a difference is smaller than the sum of the two individual SEs.

```r
# Overlapping CIs, significant difference
ci_a <- t.test(group_a)$conf.int
ci_b <- t.test(group_b)$conf.int
diff_test <- t.test(group_a, group_b)

cat("Group A 95% CI:", round(ci_a[1], 1), "to", round(ci_a[2], 1), "\n")
```

```
Group A 95% CI: 48.6 to 52.6
```

```r
cat("Group B 95% CI:", round(ci_b[1], 1), "to", round(ci_b[2], 1), "\n")
```

```
Group B 95% CI: 53.3 to 57.4
```

```r
cat("Intervals overlap?", ci_a[2] > ci_b[1], "\n")
```

```
Intervals overlap? FALSE
```

```r
cat("Difference p-value:", round(diff_test$p.value, 4))
```

```
Difference p-value: 0.0012
```

Pitfall 4: Multiple testing. Running 20 tests at $\alpha = 0.05$ produces one false-positive on average even when nothing is happening. Pre-register, or at least disclose, all comparisons. The "shoe size predicts voting" finding in a 20-variable fishing expedition is almost certainly noise.

Pitfall 5: Confusing CIs with the spread of future estimates. A 95% CI describes plausible values of the parameter, not the distribution of future point estimates. Future replications will cluster around the truth with their own (narrower) sampling distribution. More than 95% of replication estimates typically fall inside the original CI, not exactly 95%.

Pitfall 6: Letting a tight interval become a causal claim. A narrow confidence interval says the *association* is estimated precisely. It says nothing about whether the association is causal. A regression of income on years of education might return a coefficient of 3,200 with a 95% CI of [2,800, 3,600] — an impressively precise estimate — and still tell us almost nothing about what an extra year of schooling would do to a given person's earnings, because family background, ability, and geography all push education and income in the same direction. Precision and identification are different problems, and more data fixes only the first. The vocabulary should track this: write "associated with," not "causes" or "leads to," unless the design (random assignment, or a credible natural experiment) earns the stronger verb.

> **Common Mistake: "Absence of evidence is evidence of absence"**
>
> A study with p = 0.15 and a wide CI does not prove no effect exists. It proves the study lacked precision to detect one. If the CI ranges from -2 to $+10$, effects up to 10 points remain plausible. Always check the CI before concluding "no effect."

3 Using AI for This Material

> **Where AI Helps This Week**
>
> - Explaining what a p-value is (and what it is not) in the abstract, including the misinterpretations to avoid.
> - Writing `t.test()` and `lm()` syntax for a stated two-group comparison and walking through the output.
> - Describing what the t-distribution is doing and why we use it instead of the normal at small sample sizes.

Where AI Gets This Wrong This Week

- Interpreting a marginal p-value (e.g., $p = 0.06$) in context depends on prior evidence, sample size, multiple testing, and stakes — AI defaults to "not significant" or "marginally significant" without that context.
- Distinguishing statistical from practical significance for your specific finding requires knowing the substantive scale: a 0.2-point difference can be statistically significant with $N = 50{,}000$ and substantively trivial.
- It will not flag when you have implicitly run many tests and a single $p < 0.05$ is what you would expect by chance.

A prompt that works for this week:

> Write R code to run a two-sample t-test comparing `turnout` between `treatment` and `control` groups in a data frame called `gotv`, and explain what each piece of the output means.

A prompt that misleads:

> My p-value is 0.06. Is my finding significant?

Why it misleads: the question treats significance as a yes/no property of a number, but the right answer depends on prior evidence, the substantive size of the effect, and how many tests you ran — none of which AI can know from the p-value alone.

Where AI Helps This Week

- Computing a confidence interval from a given mean, standard error, and sample size, and showing the arithmetic step by step.
- Explaining the algebraic equivalence between a 95% CI excluding zero and a two-sided $p < 0.05$.
- Translating a CI into a one-sentence interpretation suitable for a non-technical audience.

Where AI Gets This Wrong This Week

- Deciding what confidence level is appropriate for a specific decision is contextual: 95% is convention, not law — high-stakes regulatory decisions may justify 99%, exploratory work may justify 90%.
- Interpreting a CI that contains zero as "no effect" is wrong: the data are consistent with a range of effects *including* zero, not with zero specifically.

- Judging whether two overlapping CIs imply no significant difference is unreliable — they often do not, and AI will commonly answer this question the wrong way.

A prompt that works for this week:

> I have a sample mean of 52.3, a standard error of 1.4, and $n = 400$. Compute a 95% confidence interval and write one sentence interpreting it for a policy audience.

A prompt that misleads:

> My 95% confidence interval is [-0.5, 3.2]. Does this mean my treatment had no effect?

Why it misleads: a CI containing zero means the data are *consistent with* a zero effect, not that the effect is zero — AI tends to flatten this into a clean "no effect" verdict that overstates what the interval actually says.

3.1 R Functions Reference

Task	Code
One-sample t-test	`t.test(x, mu = value)`
Two-sample t-test	`t.test(y ~ group, data = df)`
Change confidence level	`t.test(..., conf.level = 0.99)`
Regression CIs	`confint(model)`
Tidy output with CIs	`tidy(model, conf.int = TRUE)`
Whole-model summary	`glance(model)`
t critical value	`qt(0.975, df = n - 2)`
p-value from t-statistic	`2 * (1 - pt(t_value, df))`

4 R Functions Reference

Task	Code
Standard error of a mean	`sd(x) / sqrt(length(x))`
Standard error of a proportion	`sqrt(p * (1 - p) / n)`
One-sample t-test	`t.test(x, mu = value)`
Two-sample t-test	`t.test(y ~ group, data = df)`
Regression with t-tests for coefficients	`lm(y ~ x1 + x2, data = df)`
Tidy test or model output	`tidy(test_result)`
Critical value (t, two-sided)	`qt(0.975, df = n - 1)`
Sampling distribution by simulation	`replicate(n_sim, ...)`

5 Practice Problems

Conceptual Questions

First, in your own words, explain the difference between standard deviation and standard error. Give a political example in which the two answer different questions.

Second, a poll of 600 voters finds 54% support for a ballot initiative. Calculate the standard error and the 95% margin of error. If the true population value were 50%, how surprised should you be by 54%?

Third, a researcher runs a two-sample t-test comparing turnout between treatment and control in a get-out-the-vote experiment with $n = 200$ per group. The result is $\hat{\beta} = 3$ percentage points with $SE = 2.5$ and $p = 0.23$. Did the campaign work?

Fourth, a regression of vote share on campaign spending and incumbency yields a spending coefficient of 0.025 with $SE = 0.005$. Compute the t-statistic by hand and interpret what the test is asking.

Fifth, a graduate student tests 15 different campaign messages on small focus groups and finds that one is significant at $p = 0.04$. What concern should you raise, and what would you ask the student to do before drawing conclusions?

For Further Study

The framework in this chapter generalizes in several directions. Confidence intervals, introduced in the second half of this chapter, provide a range of plausible values for parameters rather than a single decision. Power analysis lets researchers compute the sample size needed to detect effects of a given size with a given probability. Multiple-comparison corrections like Bonferroni and false discovery rate provide formal adjustments when many tests are run together. Bayesian inference offers an alternative framework in which we directly compute the probability of hypotheses given data, sidestepping some of the interpretive traps of p-values. Each of these extends the ideas developed in this chapter without replacing them.

Conceptual Review

First, a 95% CI for a regression coefficient is [0.5, 3.2]. What do you know about the p-value, whether the coefficient is significant at $\alpha = 0.05$, and how a 99% CI would compare in width?

Second, two studies estimate the same effect. Study A (n = 50) reports $\hat{\beta} = 5.0$ with 95% CI [0.8, 9.2]. Study B (n = 500) reports $\hat{\beta} = 2.0$ with 95% CI [1.4, 2.6]. Which study provides stronger evidence of an effect? Which suggests a larger effect? What might explain the difference?

Third, a researcher reports no significant effect with p = 0.15 in a study of 25 participants. What additional information would help interpret this finding? What alternative explanations should you consider before concluding ”no effect”?

Fourth, write the correct interpretation of a 95% confidence interval in one sentence. Then write three common incorrect interpretations and explain why each is wrong.

Integrative Review

Fifth, a randomized experiment finds that mailing voters a turnout reminder increased turnout by 3 percentage points (95% CI: 0.5 to 5.5, p = 0.02, n = 1000). Write a four-sentence summary that covers: the causal claim, the effect size, the uncertainty, and a practical interpretation.

Sixth, in the regression Approval = 50 + 8(Democrat) + 12(Republican) + 0.3(Age) with Independent as the reference category, what is the predicted approval for a 40-year-old Republican? For a 40-year-old Independent?

Seventh, suppose you regress test scores on study hours and class size and find $\hat{\beta}_{study} = 3.2$ with 95% CI [2.1, 4.3], and $\hat{\beta}_{size} = -0.4$ with 95% CI [-0.9, 0.1]. Interpret each coefficient, including its significance status, and write what you can and cannot conclude about causation.

For Further Study

The methods in this course generalize. Robust standard errors handle violations of regression assumptions. Bootstrapping builds confidence intervals without distributional assumptions. Bayesian credible intervals offer a different philosophical interpretation in which probability statements about parameters become meaningful. Power analysis uses the same formulas in reverse to determine sample sizes before data collection. Difference-in-differences, instrumental variables, regression discontinuity, and matching extend causal inference to settings without random assignment. Each of these builds directly on the foundation of design, description, regression, and inference established here. If you keep going in this area, you will recognize the bones of every method as the same bones you learned this term.